

46110 Fundamentals of Aerodynamics Assignment 3

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1 Project Overview: Ingenuity Legacy

The objective of this project is to develop a conceptual aerodynamic design for a Mars drone capable of carrying a 2 kg payload. The design is based on rotor aerodynamics in hover and forward flight, airfoil selection, and optional wing integration to reduce power consumption.

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2 Task 1

2.1 First estimate of Ingenuity rotor power in hover

As a first estimate, the rotor power is evaluated using momentum theory for the induced power and a simple profile-drag power model for the blade drag. The rotor system is modeled as two coaxial rotors with equal thrust sharing, and the lower rotor is assumed to operate in the wake of the upper rotor.

2.1.1 Given data

The following values are used:

$$m = 1.8 \text{ kg}, \quad g = 3.73 \text{ m/s}^2, \quad \rho = 0.020 \text{ kg/m}^3 \quad (1)$$

The mass and rotor diameter are taken from the task description, while the Martian gravity and average surface atmospheric density are taken from NASA data [3].

$$D = 1.2 \text{ m}, \quad R = \frac{D}{2} = 0.6 \text{ m}, \quad A = \pi R^2 = \pi(0.6)^2 = 1.131 \text{ m}^2 \quad (2)$$

$$\Omega = 2800 \text{ rpm} = 2800 \cdot \frac{2\pi}{60} = 293.2 \text{ rad/s} \quad (3)$$

$$N_b = 2, \quad C_{d0} = 0.02, \quad k = 1.15 \quad (4)$$

The number of blades and blade drag coefficient follow the assumptions given in the assignment. The correction factor k is taken from the lecture slides.

2.1.2 Effective chord estimation

The Ingenuity rotor blades are highly tapered, as shown in the published geometry data [2]. Therefore, using a constant chord is not appropriate.

The reported blade area in the NASA geometry report is the total blade area of the coaxial system, i.e. both rotors combined [2]. Hence, the blade area for one rotor is

$$S_{b,\text{rotor}} = \frac{0.170}{2} = 0.085 \text{ m}^2 \quad (5)$$

An effective mean chord for one rotor is then computed from the solidity

$$\bar{c} = \frac{S_{b,\text{rotor}}}{N_b R} \quad (6)$$

which gives

$$\bar{c} = \frac{0.085}{2 \cdot 0.6} = 0.0708 \text{ m} \quad (7)$$

This value is physically consistent with the tabulated chord distribution, where the maximum chord is approximately 0.122 m [2].

2.1.3 Thrust distribution

The total thrust required in hover is taken equal to the weight:

$$T = mg = 1.8 \cdot 3.73 = 6.714 \text{ N} \quad (8)$$

As a first approximation, the thrust is assumed to be shared equally between the two rotors:

$$T_u = T_l = \frac{T}{2} = 3.357 \text{ N} \quad (9)$$

2.1.4 Upper rotor

The upper rotor is assumed to operate in undisturbed flow. Using disk momentum theory, the induced velocity in hover is

$$v_u = \sqrt{\frac{T_u}{2\rho A}} \quad (10)$$

Substituting the values from Eqs. (2) and (9) gives

$$v_u = \sqrt{\frac{3.357}{2 \cdot 0.020 \cdot 1.131}} = 8.61 \text{ m/s} \quad (11)$$

The induced power is

$$P_{i,u} = T_u v_u \quad (12)$$

$$P_{i,u} = 3.357 \cdot 8.61 = 28.92 \text{ W} \quad (13)$$

The profile power is estimated as

$$P_0 = \frac{1}{8} \rho \bar{c} N_b C_{d0} \Omega^3 R^4 \quad (14)$$

$$P_0 = \frac{1}{8} (0.020)(0.0708)(2)(0.02)(293.2)^3 (0.6)^4 = 23.14 \text{ W} \quad (15)$$

The total upper rotor power is

$$P_u = k P_{i,u} + P_0 \quad (16)$$

$$P_u = 1.15 \cdot 28.92 + 23.14 = 56.40 \text{ W} \quad (17)$$

2.1.5 Lower rotor

The lower rotor operates in the wake of the upper rotor. The inflow is modeled as

$$V_c = fw = f2v_u \quad (18)$$

Using $f = 0.5$, this becomes

$$V_c = v_u = 8.61 \text{ m/s} \quad (19)$$

The thrust equation becomes

$$T_l = 2\rho A(V_c + v_{i,l})v_{i,l} \quad (20)$$

which gives

$$v_{i,l} = -\frac{V_c}{2} + \sqrt{\left(\frac{V_c}{2}\right)^2 + \frac{T_l}{2\rho A}} \quad (21)$$

$$v_{i,l} = 5.32 \text{ m/s} \quad (22)$$

The induced power is

$$P_{i,l} = T_l v_{i,l} \quad (23)$$

$$P_{i,l} = 3.357 \cdot 5.32 = 17.87 \text{ W} \quad (24)$$

and the total lower rotor power is

$$P_l = kP_{i,l} + P_0 \quad (25)$$

$$P_l = 1.15 \cdot 17.87 + 23.14 = 43.70 \text{ W} \quad (26)$$

2.1.6 Total rotor power

$$P_{\text{tot}} = P_u + P_l \quad (27)$$

$$P_{\text{tot}} = 56.40 + 43.70 = 100.10 \text{ W} \quad (28)$$

Thus, the first estimate of the total rotor power is

$$\boxed{P_{\text{tot}} \approx 100 \text{ W}} \quad (29)$$

which corresponds to approximately

$$\boxed{P_{\text{rotor}} \approx 50 \text{ W per rotor}} \quad (30)$$

2.1.7 Comment

The estimate is based on a simplified coaxial rotor model with equal thrust sharing and a constant effective chord. The real Ingenuity rotor exhibits strong taper and twist, and the result should therefore be interpreted as a first-order estimate. The obtained rotor power is significantly lower than the total in-flight power reported for Ingenuity, which indicates that the simplified model likely underpredicts the true aerodynamic power requirement and does not account for the full complexity of coaxial rotor interaction and other system losses.

3 Task 2

3.1 Geometry and Mass

First, we need to derive an equation to describe power during hover as a function of number of rotors and radius of blades. As we're basing this off our initial setup, for now chord length will stay constant.

Starting with the effective chord estimation, we have to redefine our blade area to be a function of blade radius. Since we know there's a blade area of $0.085m^2$ for a rotor of two blades with a diameter of $1.2m$, we can assume:

$$S_{b,blade} = \frac{0.085}{1.2}m = 0.0708m^2/m. \quad (31)$$

Then we can get our effective mean chord for one rotor as a function of radius.

$$\bar{c} = \frac{S_{b,blade}2R}{N_b R} = 2 \frac{S_{b,blade}}{N_b} = \frac{0.14167}{N_b} \quad (32)$$

For our mass, we need to make a few assumptions, as the scaled up version will need to change from the original Ingenuity design. We know that the total rotor weight is $70g$, so we can get the following for hypothetical blade weight:

$$m_{blade} = \frac{70g}{N_{rotors}N_b R_{Ingenuity}} = \frac{70}{2 * 2 * 0.6} = 29.167g/m \quad (33)$$

Next, we'll assume that our battery pack's capacity has increased to $20 Wh$, and the mass has increased to $500g$. The computer and other components will slightly scale to $1kg$. For the motors, we need to do more math, again based off Ingenuity as our example:

$$m_{motor} = \frac{250g * P_{new}}{N_{rotors}P_{Ingenuity}} = \frac{250}{2 * 50} = 2.5g/W \quad (34)$$

Finally, our mass of the new fuselage has to scale with everything's mass increasing, which we can derive a simple formula for here:

$$m_{struct} = m_{no-struct} * \frac{m_{struct-Ingenuity}}{m_{no-struct-Ingenuity}} = m_{no-struct} * \frac{300}{1500} = 0.2m_{no-struct} \quad (35)$$

The masses are collected in the table here:

Weight	Ingenuity	Proposal
Rotors	$70g$	$29.167g/m$
Motors	$250g$	$2.5P_{tot}g$
Structure	$300g$	$0.2m_{no-struct}g$
Battery	$280g$	$500g$
Computer	$900g$	$1000g$

Table 1: Compared Mass of Proposed Design

This leaves us with the following equation for m_{new} :

$$m_{new}(g) = 1.2(1500 + 29.167 * R * N_b + 2.5 * P_{tot}) \quad (36)$$

3.2 Thrust

The thrust in hover can again be derived, but now as a function of radius and Power as well:

$$T = mg = m_{tot} * 3.73 \quad (37)$$

We then assume that thrust is shared equally among all rotors:

$$T_{rotor} = \frac{3.73m_{tot}}{N_{rotors}} \quad (38)$$

3.3 Disk Momentum Application

Similar to Task 1, we're starting with finding our induced velocity in hover.

$$v_i = \sqrt{\frac{3.73m_{tot}}{N_{rotors}2\rho A}} \quad (39)$$

This now gives us induced power:

$$P_i = \frac{3.73m_{tot}}{N_{rotors}} * \sqrt{\frac{3.73m_{tot}}{N_{rotors}2\rho A}} = \sqrt{\frac{51.895m_{tot}^3}{2N_{rotors}^3\rho A}} \quad (40)$$

Assuming the same assumptions in task 1 (2 blades/rotor, blade drag coefficient, k factor) we can find the profile power, with the additional assumption of the same rotor speed Ω .

$$P_0 = \frac{1}{8}\rho\bar{c}N_bC_{d0}\Omega^3R^4 \quad (41)$$

$$P_0 = \frac{1}{8}(0.020)\left(\frac{0.14167}{N_b}\right)N_b(0.02)(293.2)^3R^4 = 178.72R^4 \quad (42)$$

Finally, by adding our power equations and multiplying by our number of rotors we can get an equation for total power depending on number of blades and blade length:

$$P_{tot} = \left(\sqrt{\frac{51.895m_{tot}^3}{2N_{rotors}^3\rho\pi R^2}} + 178.72R^4\right)N_{rotors} \quad (43)$$

From this point we can create a system of equations with out equation for m_{tot} , and solve for a range of r values at various values for N_{rotors} . First is a wide range of values to get the overall trend, that as rotors increase efficiency increases, but eventually for marginal gains.

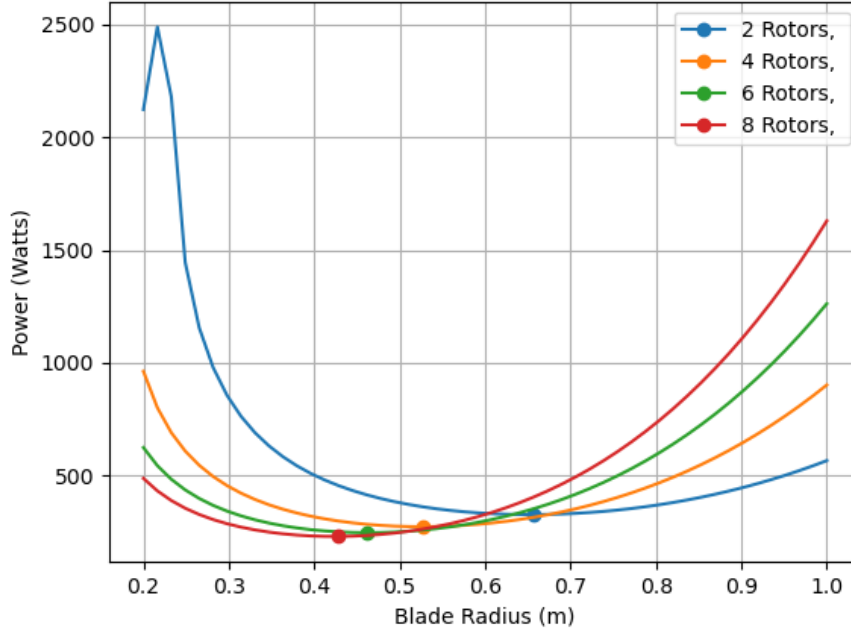


Figure 1: Power as a Function of Radius and N_{rotors}

The following graph shows the output from solving this system, with the minima labeled, and from this we can get our design constraints and settle on 4 blades, from the reduced overall power.

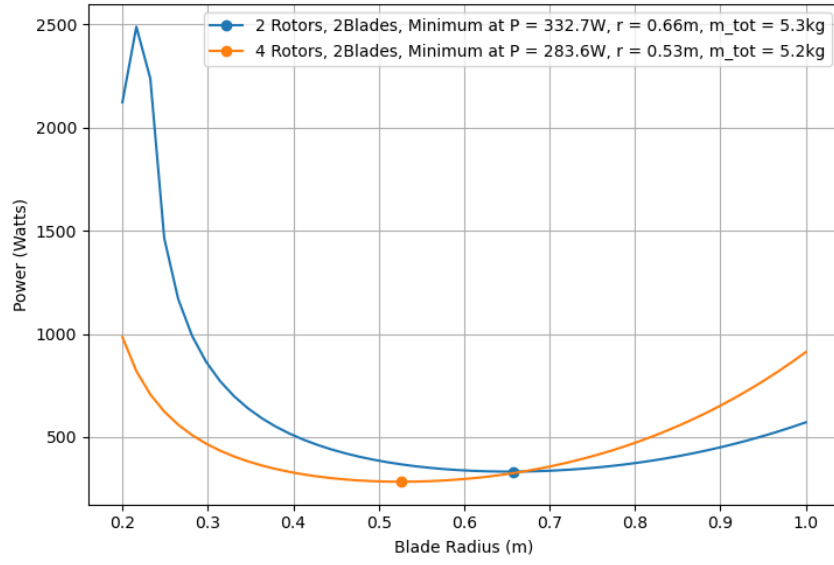


Figure 2: Power as a Function of Radius and N_{rotors}

Displaying our data in a table with flight times, we get the following:

Variable	Two Rotors	Four Rotors
Radius	0.64m	0.53m
Blades	2	2
Structure	2.40kg	2.77kg
Power Consumption	328.2W	274.6W
Flight Time (min)	3.66	4.7

Table 2: Compared Mass of Proposed Design

3.4 Task 3.1 and Task 3.2

The objective of this task is to increase the flight time by replacing the payload with additional batteries.

The drone configuration selected in Task 2 is kept fixed:

$$N_{\text{rotors}} = 4, \quad N_b = 2, \quad R = 0.53 \text{ m} \quad (44)$$

The payload mass of 2 kg is replaced by battery cells. Each battery cell has:

$$m_{\text{cell}} = 0.047 \text{ kg}, \quad E_{\text{cell}} = \frac{10}{6} \text{ Wh} \quad (45)$$

This means that the maximum number of batteries that can be added is:

$$N_{\text{extra}} \leq \frac{2}{0.047} \approx 42 \quad (46)$$

For each possible number of batteries, the total mass of the drone is recalculated as:

$$m_{\text{tot}} = 1.2 m_{\text{no-struct}} \quad (47)$$

The required hover power is computed using:

$$P_{\text{tot}} = \left(\sqrt{\frac{51.895 m_{\text{tot}}^3}{2 N_{\text{rotors}}^3 \rho \pi R^2}} + 178.72 R^4 \right) N_{\text{rotors}} \quad (48)$$

The total available energy is:

$$E_{\text{tot}} = 20 + N_{\text{extra}} \cdot \frac{10}{6} \quad (49)$$

The flight time is then calculated as:

$$t_{\text{flight}} = 60 \frac{E_{\text{tot}}}{P_{\text{tot}}} \quad (50)$$

These equations are needed because the flight time depends on two competing effects: adding batteries increases the available energy, but it also increases the mass, which leads to a higher power requirement.

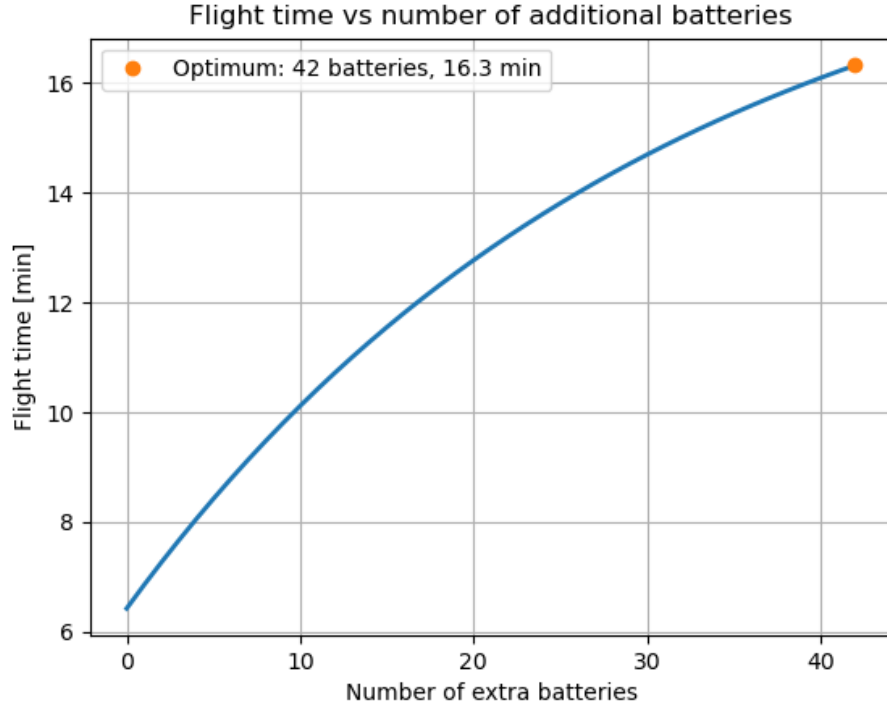


Figure 3: Flight time as a function of the number of additional batteries.

From the results, it can be seen that the flight time increases steadily as more batteries are added. This is because the additional energy has a stronger effect than the increase in power required to carry the extra weight. In this configuration, the maximum flight time is reached at the payload limit. The optimal number of batteries is therefore:

$$N_{\text{extra}} = 42 \quad (51)$$

which corresponds to a flight time of approximately:

$$t_{\text{flight}} \approx 16.3 \text{ min} \quad (52)$$

3.5 Comparison with original configuration

For comparison, the flight time is also computed using only the original battery pack, without any additional batteries. It corresponds to the starting point of our plot : 0.5 Kg of original batteries with a flight time of 13.1 minutes (taken from the graph) This result highlights the significant improvement obtained with additional batteries.

3.5.1 Discussion

The results highlight the balance between energy and weight.

At first, adding batteries is very beneficial because it increases the available energy significantly. At the same time, the increase in mass is still relatively small, so the power requirement does not increase too much.

As more batteries are added, the mass becomes more important and the power required increases. In many cases, this would eventually reduce the flight after a certain point.

The best solution is simply to use as many batteries as the drone can carry.

In short, replacing the payload with batteries improves the flight time, and the optimal solution is obtained at the maximum allowable number of batteries.

3.6 Task 3.3 Limit check

The maximum number of batteries that can be carried is limited by the payload mass:

$$N_{\text{max}} = \left\lfloor \frac{2}{0.047} \right\rfloor = 42 \quad (53)$$

From the optimization:

$$N_{\text{extra,opt}} = 42 \quad (54)$$

Since:

$$N_{\text{extra,opt}} = N_{\text{max}} \quad (55)$$

the optimal solution is exactly at the payload limit.

The total battery mass is:

$$m_{\text{batteries,tot}} = 0.047 \cdot 42 = 1.97 \text{ kg} < 2 \text{ kg} \quad (56)$$

This shows that the payload constraint is satisfied.

For completeness, if the optimal number of batteries had been higher than the payload limit, the final design would have been constrained by N_{max} , and the flight time would have been evaluated at that point.

The final result is:

$$t_{\text{flight}} \approx 16.3 \text{ min} \quad (57)$$

4 Task 4

4.1 Atmospheric properties and Reynolds number estimation

In order to select a suitable airfoil for the rotor and potential wing design, it is first necessary to estimate the Reynolds number under Martian conditions and compare it with Earth.

4.1.1 Atmospheric properties

The Martian atmosphere is much thinner than Earth's and is mainly composed of carbon dioxide. For the present analysis, the following representative values are used (taken from NASA data [3]):

$$\rho_{\text{Mars}} = 0.020 \text{ kg/m}^3, \quad \mu_{\text{Mars}} = 1.05 \times 10^{-5} \text{ Pa} \cdot \text{s} \quad (58)$$

For comparison, standard sea-level conditions on Earth are:

$$\rho_{\text{Earth}} = 1.225 \text{ kg/m}^3, \quad \mu_{\text{Earth}} = 1.789 \times 10^{-5} \text{ Pa} \cdot \text{s} \quad (59)$$

4.1.2 Comparison of Reynolds number

The Reynolds number is defined as:

$$Re = \frac{\rho V c}{\mu} \quad (60)$$

For the same velocity and chord length, the ratio between Mars and Earth Reynolds numbers is:

$$\frac{Re_{\text{Mars}}}{Re_{\text{Earth}}} = \frac{\rho_{\text{Mars}}/\mu_{\text{Mars}}}{\rho_{\text{Earth}}/\mu_{\text{Earth}}} \quad (61)$$

Substituting the values gives:

$$\frac{Re_{\text{Mars}}}{Re_{\text{Earth}}} \approx 0.028 \quad (62)$$

This shows that the Reynolds number on Mars is approximately 36 times smaller than on Earth for the same flow conditions. This is a key result, as it implies that the flow operates in a very low Reynolds number regime, which significantly affects airfoil performance.

4.1.3 Reynolds number at $r = 0.75R$

To estimate a representative Reynolds number for the rotor, the flow conditions are evaluated at $r = 0.75R$, which is commonly used as a reference location along the blade. A more complete expression for the local relative velocity includes axial flow:

$$V_{\text{rel}} = \sqrt{(\Omega r)^2 + (V_c + v_i)^2} \quad (63)$$

However, since $\Omega r \gg (V_c + v_i)$, the tangential velocity dominates and:

$$V_{\text{rel}} \approx \Omega r \quad (64)$$

Using the rotor speed:

$$\Omega = 2800 \text{ rpm} = 293.2 \text{ rad/s} \quad (65)$$

and the Ingenuity rotor radius:

$$R = 0.605 \text{ m}, \quad r = 0.75R = 0.454 \text{ m} \quad (66)$$

the local velocity becomes:

$$V_{0.75R} = \Omega r = 293.2 \times 0.454 \approx 133 \text{ m/s} \quad (67)$$

The chord at $r = 0.75R$ is approximated as being similar to the effective mean chord obtained in Task 1:

$$c_{0.75R} \approx \bar{c} = 0.0708 \text{ m} \approx 0.071 \text{ m} \quad (68)$$

This approximation is considered sufficient for the Reynolds number estimation, since the objective is to determine the correct order of magnitude.

The Reynolds number is therefore:

$$Re_{0.75R} = \frac{\rho_{\text{Mars}} V_{0.75R} c_{0.75R}}{\mu_{\text{Mars}}} = \frac{(0.020)(133)(0.071)}{1.05 \times 10^{-5}} \approx 1.8 \times 10^4 \quad (69)$$

4.1.4 Comment

The obtained Reynolds number is on the order of 10^4 , which is significantly lower than typical values for aerodynamic applications on Earth. This confirms that the rotor operates in a low-Reynolds-number regime, where viscous effects are dominant. Consequently, airfoil selection must focus on profiles that perform well at low Reynolds numbers.

4.2 Airfoil selection

4.2.1 Selection methodology

The selection of candidate airfoils was based on a filtering process.

Given the estimated Reynolds number of $Re \approx 1.8 \times 10^4$, the aerodynamic problem lies in the very-low-Reynolds-number regime. In this regime, viscous effects dominate the flow behavior, and classical airfoils are known to suffer from early separation and performance degradation.

Based on literature sources, such as Mars rotorcraft studies and low-Reynolds-number aerodynamic analyses, the selection was therefore restricted to airfoils specifically designed or demonstrated to perform well under these conditions. In particular, thin cambered profiles and Mars-specific airfoils have been identified as the most suitable configurations.

Instead of analyzing a large number of airfoils, a limited set of representative candidates was chosen in order to explore different aerodynamic behaviors. The selected airfoils (taken from Nasa report [3]) represent three key categories relevant to the problem:

- ROAMX-0201: a Mars-optimized airfoil developed for rotorcraft applications at very low Reynolds numbers,
- CLF5605: the airfoil used on the Ingenuity helicopter, providing a flight-proven baseline configuration,
- 5% circular-arc plate: a thin cambered profile representative of simple geometries known to perform well in very-low-Reynolds-number conditions.

This selection approach allows for a structured and meaningful comparison, while remaining consistent with the literature and the specific aerodynamic regime of the problem.

4.2.2 Airfoil geometries

The geometries of the selected airfoils were obtained from literature sources (Nasa report [4]) related to Mars rotorcraft design.

The ROAMX-0201 airfoil is a thin, highly cambered profile specifically designed to maximize lift performance in the Martian atmosphere. Its geometry is optimized for low-density and low-Reynolds-number conditions.

The CLF5605 airfoil is the profile used on the Ingenuity helicopter blades. It represents a practical compromise between aerodynamic performance and structural constraints, and serves as a validated reference configuration.

The 5% circular-arc plate is a simplified airfoil geometry characterized by a thin profile and moderate camber. Such profiles are known to perform well at very low Reynolds numbers due to their reduced sensitivity to separation and their ability to maintain lift in laminar flow conditions.

4.2.3 Aerodynamic performance

Due to the limited availability of tabulated polar data at very low Reynolds numbers, the aerodynamic performance of the selected airfoils was assessed using data extracted from literature sources. The available data correspond to conditions close to the operating regime of interest and are therefore considered representative for the present analysis. It should be noted that the data were extracted from literature figures and therefore represent approximate trends rather than exact quantitative values.

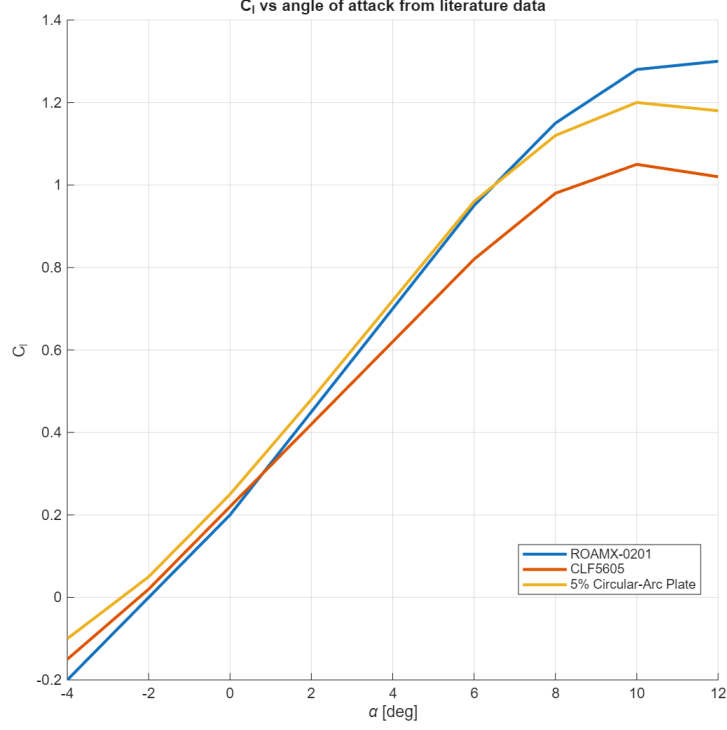


Figure 4: C_l as a function of angle of attack for the selected airfoils based on literature data.

The C_l versus angle of attack curves show that the ROAMX-0201 airfoil achieves the highest lift coefficients, reaching values above $C_l \approx 1.3$. The lift curve remains smooth over the considered range, indicating stable aerodynamic behavior.

The 5% circular-arc plate also exhibits strong lift generation, with slightly lower maximum lift coefficients. Its behavior is relatively linear and stable, making it a robust candidate for low-Reynolds-number operation.

The CLF5605 airfoil shows lower lift levels compared to the other two profiles. While its behavior is stable, it requires higher angles of attack to reach comparable lift values.

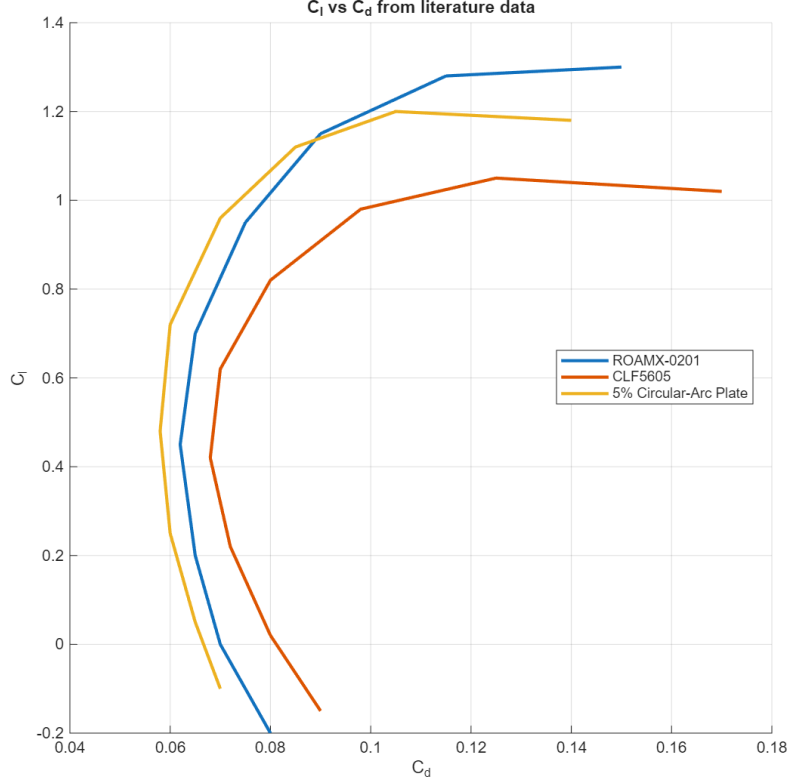


Figure 5: C_l as a function of C_d for the selected airfoils based on literature data.

The C_l versus C_d curves highlight the aerodynamic characteristics of the airfoils. The ROAMX-0201 airfoil achieves higher lift coefficients across the operating range, while maintaining drag levels comparable to the other profiles.

The 5% circular-arc plate shows competitive performance and relatively low drag in the moderate lift range, confirming its suitability as a robust low-Reynolds-number solution.

The CLF5605 airfoil exhibits higher drag for a given lift level, resulting in lower overall aerodynamic performance compared to the other candidates.

4.2.4 Turbulator consideration

A turbulator is a small roughness element placed on the airfoil surface to force an earlier transition from laminar to turbulent boundary layer. At very low Reynolds numbers, the boundary layer tends to remain laminar over a large portion of the airfoil, which can lead to early flow separation and the formation of laminar separation bubbles.

By promoting earlier transition, a turbulator can delay separation and improve lift behavior. However, this also increases skin-friction drag, and therefore its use must be carefully evaluated.

For the ROAMX-0201 airfoil, which already exhibits strong low-Reynolds-number performance, the addition of a turbulator could further stabilize the flow. However, its optimized design suggests that it already accounts for such effects, and the benefit may therefore be limited.

For the CLF5605 airfoil, which shows higher drag and lower lift performance, the use of a turbulator is unlikely to provide significant improvement and may further increase drag.

For the 5% circular-arc plate, which relies on simple geometry and laminar flow behavior, the addition of a turbulator could help delay separation. However, it would also reduce the advantages associated with its low-drag characteristics.

In the present conceptual design, no turbulator is included, but its potential use should be considered in a

more detailed design phase. At very low Reynolds numbers, however, the use of a turbulator is often a trade-off between improved flow attachment and increased drag, and its effectiveness depends strongly on the specific airfoil geometry.

4.2.5 Final airfoil selection

Based on the comparison, the ROAMX-0201 airfoil is selected as the most suitable candidate.

While the 5% circular-arc plate provides robust and stable performance, its maximum lift coefficient remains lower than that of ROAMX-0201. The CLF5605 airfoil, although flight-proven, exhibits lower lift and higher drag compared to the other candidates.

The ROAMX-0201 airfoil offers the highest lift capability while maintaining smooth and stable aerodynamic behavior, making it particularly well suited for very-low-Reynolds-number conditions. This makes it the most appropriate choice for the present Martian rotor application.

5 Task 5

The optimal, idealized twist distribution was found by implementing a simple BEM model for the rotor. By taking our known values for C_l and C_d as a function of α we can go along the blade to find the optimal twist.

5.1 BEM Method

For each element along the blade, our method is the same. We set up an equation with the following constraints, and solved the optimization problem. This is following the method described by the Rotor-Aerodynamics chapter [7], BEM chapter of Hansen, Aerodynamics of Wind Turbines [1], and lecture 11 *Blade-element momentum (BEM) theory for rotors* [6].

Variable	Value	Source
r_{blade}	$0.53m$	Task 2
$n_{elements}$	50	Iteration
N_b	2	Task 2
$T_{target(rotor)}$	$4.27N$	Task 2
$r_{hubFrac}$	0.05	Guess

Table 3: Constraining Variables

Variable	Bounds	Description
$C_l(tip)$	$(0.005, 0.040)m$	Tip Chord bounds
θ_{pitch}	$(-10, 15)deg$	Collective Pitch Bounds
α_{twist}	$(0, 10)deg$	Blade Twist Bounds
Ω	$(200, 1500)rad/s$	Blade Velocity

Table 4: Optimized Variables

How our implementation varies a bit is the inclusion of factors such as $r_{hubFrac}$ which represents the unusable space near the hub in which either the blade connection or the geometry of our rotor would prevent usable blade radius, here guessed to be $0.15r_{blade}$, or about $9mm$. Additionally, the geometry bound on the tip represents what could reasonably be physically manufactured for our system. Finally, Prandtl tip loss is used with the Glauert derivation ([1],[6]), and there is no hub loss considered.

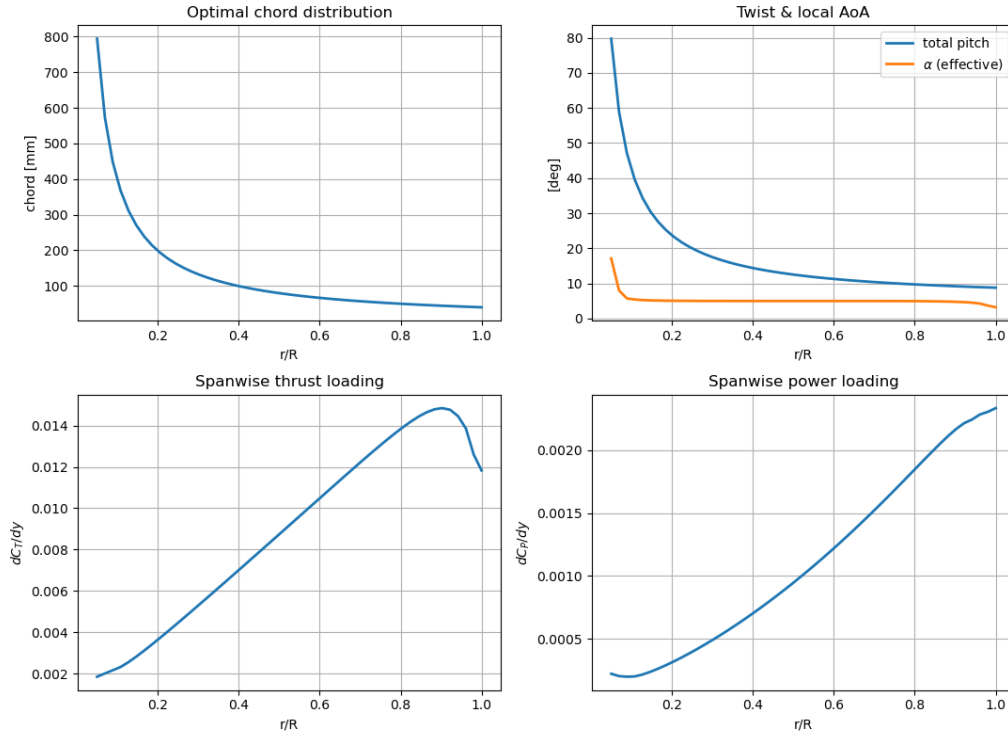


Figure 6: Optimized Airfoil Characteristics

5.2 Output Variables

From `Scipy.Optimize-Minimize` [8] using the *Nelder-Mead* method, we can solve our BEM system for our drone’s twist, pitch, thrust, and power.

Variable	Task 2	BEM Model
Power (Total)	274.6W	499W
Thrust (Rotor)	4.77N	4.77N

Table 5: BEM model and Task 2 Results for Optimized Design

Our overall power is significantly higher than what we found in Task 2. There’s a few factors that could contribute to this, being the higher accuracy of losses, such as the inclusion of the hub’s reduction of lifting area on our blade, the more accurate (and less optimal) modeling of actual blade aerodynamics, as well as Prandtl tip losses. If we ”turn off” tip loss effects, we get a per-rotor power of 100W, giving around a decent 20% reduction in power consumption.

It was useful to have the model in Task 2 to get our design constraints, such as initial radius and number of blades/rotors, but a continuation of this report could include using our BEM model to recalculate other potential rotor/blade configurations, as well as finding our new theoretical maximum flight time Task 3.

6 Task 6 - Forward Flight Analysis

6.1 Overview

Task 6 evaluates whether adding a fixed wing to the quadcopter from Tasks 2 and 5 reduces total rotor power during forward flight at $V = 10 \text{ m/s}$ on Mars. The forward flight model follows the momentum theory framework from Lecture 12, consistent with the hover analysis in Task 2; the BEM-based power from Task 5 would raise the absolute baseline but would not alter the overall conclusion, as the wing benefit scales with thrust sensitivity rather than the choice of rotor power model. The hover power from Task 2 is $P_{\text{hover}} = 273.9 \text{ W}$, used here as the baseline reference. The aircraft parameters are given in Table 6.

Table 6: Quadcopter design parameters (from Tasks 2 and 5).

Parameter	Symbol	Value	Unit
Total mass (incl. 2 kg payload)	m	4.6	kg
Number of rotors	N_{rot}	4	—
Rotor radius	R	0.53	m
Blades per rotor	N_b	2	—
Angular velocity	Ω	293.2	rad/s
Mean blade chord	$\bar{c}$	0.071	m
Blade profile drag coefficient	C_{d0}	0.02	—
Induced power correction factor	κ	1.15	—
Mars air density	ρ	0.020	kg/m ³
Mars gravitational acceleration	g	3.73	m/s ²

6.2 Forward Flight Mechanics

6.2.1 Force Balance

In steady level forward flight the rotor disk tilts forward by angle β , so that the horizontal thrust component balances the total drag D while the vertical component supports the effective weight $W_{\text{eff}} = W - L_w$ (aircraft weight reduced by any wing lift L_w). From Lecture 12:

$$T \sin \beta = D, \quad T \cos \beta = W_{\text{eff}}, \quad (70)$$

giving

$$\beta = \arctan\left(\frac{D}{W_{\text{eff}}}\right), \quad T = \sqrt{W_{\text{eff}}^2 + D^2}. \quad (71)$$

The total drag is the sum of fuselage body drag and wing drag:

$$D = q A_b C_{D,b} + q S C_{D,w}, \quad q = \frac{1}{2} \rho V^2, \quad (72)$$

where $A_b = 0.018 \text{ m}^2$, $C_{D,b} = 0.4$, S is the wing planform area, and $C_{D,w}$ is defined in Section 6.3. Each rotor carries $T_{\text{per}} = T/N_{\text{rot}}$.

6.2.2 Glauert Induced Velocity

From Lecture 12, the resultant velocity at the rotor disk is

$$U = \sqrt{(V \cos \beta)^2 + (V \sin \beta + v_i)^2}, \quad (73)$$

and momentum theory across the rotor plane gives the thrust as

$$T_{\text{per}} = 2\rho A U v_i. \quad (74)$$

Combining with the hover relation $v_H^2 = T_{\text{per}}/(2\rho A)$ from Task 2, the induced velocity satisfies the equation

$$v_i = \frac{v_H^2}{\sqrt{(V \cos \beta)^2 + (V \sin \beta + v_i)^2}}, \quad (75)$$

which is solved iteratively until convergence.

6.2.3 Total Power in Forward Flight

Following Lecture 12, the ideal power per rotor in forward flight is

$$P_{\text{ideal}} = T_{\text{per}}(V \sin \beta + v_i). \quad (76)$$

The profile drag power per rotor is identical to hover (Lecture 9):

$$P_0 = \frac{1}{8} \bar{c} N_b C_{d0} \Omega^3 R^4. \quad (77)$$

The total power for all rotors is then, following Lecture 12:

$$P = N_{\text{rot}}(\kappa P_{\text{ideal}} + P_0). \quad (78)$$

Without a wing the forward flight power at $V = 10 \text{ m/s}$ is $P_{\text{no wing}} = 234.4 \text{ W}$, which is 39.5 W below hover because the Glauert model (Eq. 75) predicts a lower induced velocity once the rotor translates: $v_i < v_H$ for $V > 0$.

6.3 Wing Aerodynamic Model

6.3.1 Airfoil

The ROAMX-0201 airfoil selected in Task 4 is also used for the wing. A linear fit to the Task 5 polar over the attached range ($\alpha \leq 4^\circ$) gives a 2-D lift slope $a_0 = 5.58 \text{ rad}^{-1}$ and zero-lift angle $\alpha_{L0} = -1.88^\circ$.

6.3.2 Lifting Line Theory

Wing induced drag and Oswald efficiency are computed using Prandtl's Lifting Line Theory (LLT), following the same formulation applied in Assignment 2. The circulation distribution is expanded in a Fourier series:

$$\Gamma(\theta) = 2bV \sum_{n=1}^N A_n \sin(n\theta), \quad y = -\frac{b}{2} \cos \theta. \quad (79)$$

Substituting into the LLT integral equation and evaluating at $N = 50$ collocation points $\theta_k = k\pi/(N+1)$ yields the linear system

$$\sum_{n=1}^N A_n \left[\frac{4b}{a_0 c(\theta_k)} \sin(n\theta_k) + \frac{n \sin(n\theta_k)}{\sin \theta_k} \right] = \alpha_{\text{geom}}(\theta_k) - \alpha_{L0}, \quad (80)$$

where $c(\theta_k)$ is the local chord and α_{geom} includes any geometric twist. Solving for A_n gives the global wing coefficients:

$$C_L = \pi \text{AR} A_1, \quad C_{D,i} = \pi \text{AR} \sum_{n=1}^N n A_n^2, \quad e = \frac{A_1^2}{\sum_{n=1}^N n A_n^2}. \quad (81)$$

The Oswald efficiency e equals unity only for a pure elliptic lift distribution; any deviation towards uniform chord or non-optimal twist causes $e < 1$ and increases induced drag. The total wing drag coefficient is

$$C_{D,w} = C_{d0}(C_L) + \frac{C_L^2}{\pi \text{AR} e}, \quad (82)$$

where $C_{d0}(C_L)$ is interpolated from the ROAMX-0201 polar.

6.4 Wing Shape Optimisation

6.4.1 Parameterisation

The wing is fully defined by three shape parameters. **Aspect ratio** $AR \in \{5, 7, 9\}$ is swept numerically; a higher AR lowers induced drag but increases structural weight. **Taper ratio** $\lambda = c_{\text{tip}}/c_{\text{root}}$ is swept over $\{0.3, 0.4, \dots, 1.0\}$ with a linear chord distribution:

$$c(y) = c_r \left[1 - (1 - \lambda) \frac{2|y|}{b} \right], \quad c_r = \frac{2b}{AR(1 + \lambda)}. \quad (83)$$

Geometric washout θ_{tip} is swept over $\{0^\circ, -1^\circ, -2^\circ, -3^\circ, -4^\circ\}$ with a linear twist distribution:

$$\alpha_{\text{geom}}(y) = \alpha_r + \theta_{\text{tip}} \frac{2|y|}{b}, \quad (84)$$

where a negative θ_{tip} reduces the tip angle of attack (washout), shifting the lift distribution inboard towards the elliptic ideal.

6.4.2 Optimum from LLT

For each $(AR, \lambda, \theta_{\text{tip}})$ the system (80) is solved and e computed from (81). Figure 7 shows how e varies with λ and θ_{tip} for each AR.

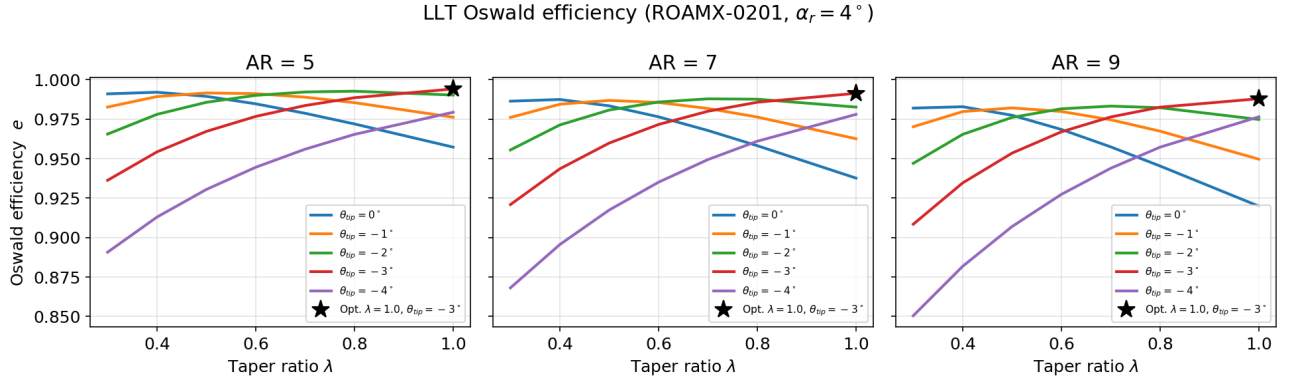


Figure 7: Oswald efficiency e from LLT as a function of taper ratio λ for five washout angles and three aspect ratios. The star marks the optimal combination per AR.

The LLT-optimal planform for all three aspect ratios is a **rectangular wing** ($\lambda = 1$) with -3° of geometric washout (Table 7).

Table 7: LLT-optimal wing shape per aspect ratio ($\alpha_r = 4^\circ$).

AR	λ^*	θ_{tip}^*	e
5	1.0	-3°	0.994
7	1.0	-3°	0.991
9	1.0	-3°	0.988

The result is physically consistent: a rectangular planform distributes chord evenly across the span so that with a moderate washout the induced-angle distribution approximates the elliptic ideal, giving $e > 0.98$. Figure 8 confirms this: the LLT spanwise $c_l(y)$ distribution is nearly uniform, close to the ideal elliptic reference.

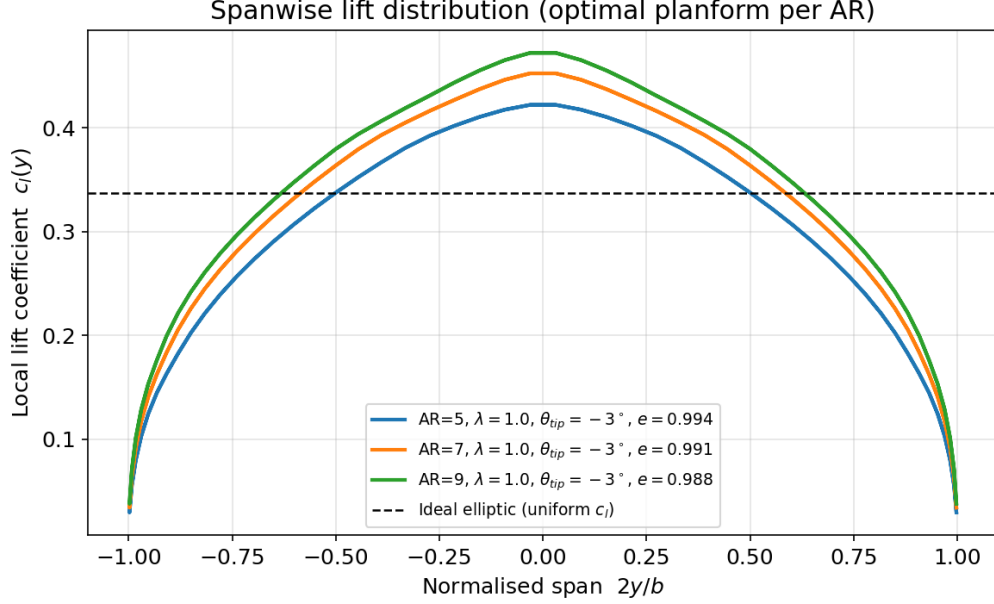


Figure 8: Spanwise lift coefficient $c_l(y)$ from LLT for the optimal planform at each AR ($\lambda = 1.0$, $\theta_{\text{tip}} = -3^\circ$). Dashed line: ideal elliptic reference (uniform c_l).

6.5 Wing Mass Model

The structural mass is estimated from the airfoil volume integrated along the span. For the ROAMX-0201 profile ($\tau = t/c = 0.05$ from the Ingenuity blade geometry) the cross-sectional area at station y is approximated as $A_{\text{sec}}(y) \approx k_s \tau c(y)^2$ with shape factor $k_s = 0.55$. Integrating over the full span:

$$V_w = 2 \int_0^{b/2} k_s \tau c(y)^2 dy = 2 k_s \tau c_r^2 \left[\frac{b}{2} - \frac{(1-\lambda)b}{2} + \frac{(1-\lambda)^2 b}{6} \right]. \quad (85)$$

For the optimal rectangular planform ($\lambda = 1$) this simplifies to $V_w = k_s \tau c_r^2 b$, and the total aircraft mass becomes $m_{\text{tot}} = m + \rho_w V_w$ with $\rho_w = 74 \text{ kg/m}^3$.

6.6 Results

6.6.1 Mars Dynamic Pressure

The dynamic pressure at the design speed is

$$q = \frac{1}{2} \rho V^2 = \frac{1}{2} \times 0.020 \times 100 = 1.0 \text{ Pa}, \quad (86)$$

approximately 60 times lower than on Earth at the same speed. Wing lift scales directly with q , making aerodynamic forces intrinsically small on Mars.

6.6.2 Wing Sweep

Figure 9 shows the total rotor power (Eq. 78) and aircraft mass as functions of wingspan b , using the LLT-optimal planform for each AR.

The maximum achievable reduction is **0.09%**. At the optimal wingspan the wing is small enough that its lift just exceeds its added weight, giving a marginal net reduction in W_{eff} and hence in rotor thrust. Beyond the optimal wingspan, structural mass grows faster than aerodynamic lift — wing mass scales with b^3 while lift scales with b^2 — so the power rises back above the no-wing baseline, as seen in Figure 10.

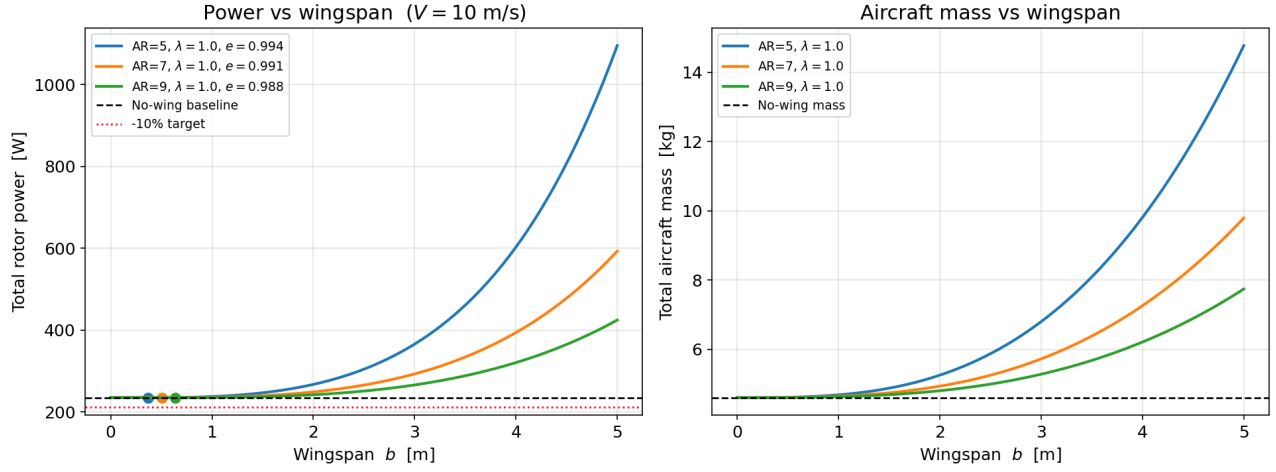


Figure 9: Left: total rotor power vs. wingspan at $V = 10$ m/s. Dashed line: no-wing baseline (234.4 W). Dotted red: 10% reduction target (211.0 W). Right: total aircraft mass vs. wingspan.

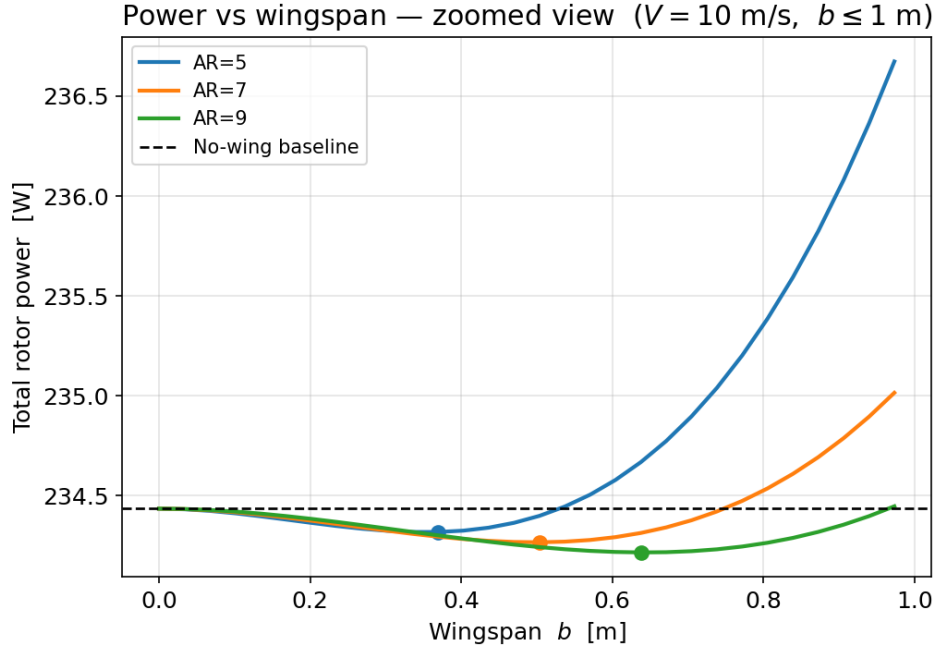


Figure 10: Zoomed view of total rotor power vs. wingspan for $b \leq 1$ m. The power dips marginally below the no-wing baseline near the optimal wingspan before rising again as wing mass dominates. Dots mark the optimal wingspan per AR.

Table 8: Optimal wingspan and power change per AR ($\rho_w = 74$ kg/m³, LLT-optimal planform).

AR	λ^*	e	b_{opt} (m)	ΔP (%)
5	1.0	0.994	0.37	-0.05
7	1.0	0.991	0.50	-0.07
9	1.0	0.988	0.64	-0.09

6.6.3 10% Reduction Feasibility

The 10% target ($P \leq 211.0 \text{ W}$) requires the wing to produce a net lift benefit large enough to reduce T_{per} and hence v_i by roughly 10%. With $\rho_w = 74 \text{ kg/m}^3$ the wing mass always outweighs its lift contribution for any wingspan. A downward density search (scanning ρ_w from 74 kg/m^3 to 0.5 kg/m^3) identifies the maximum allowable density:

$$\rho_w \leq 7.0 \text{ kg/m}^3 \text{ at } b \approx 6.77 \text{ m} \quad (\text{AR} = 9, \lambda = 1.0). \quad (87)$$

No structural composite approaches 7.0 kg/m^3 ; the 10% target is therefore not physically achievable under Martian conditions.

6.6.4 Final Design and Speed Sweep

Since the best achievable improvement is $0.09\% < 1\%$ the final design is the **bare quadcopter without a fixed wing**.

Figure 11 shows total rotor power and tilt angle β from hover to $V = 12 \text{ m/s}$, computed by applying Equations (75)–(78) at each speed.

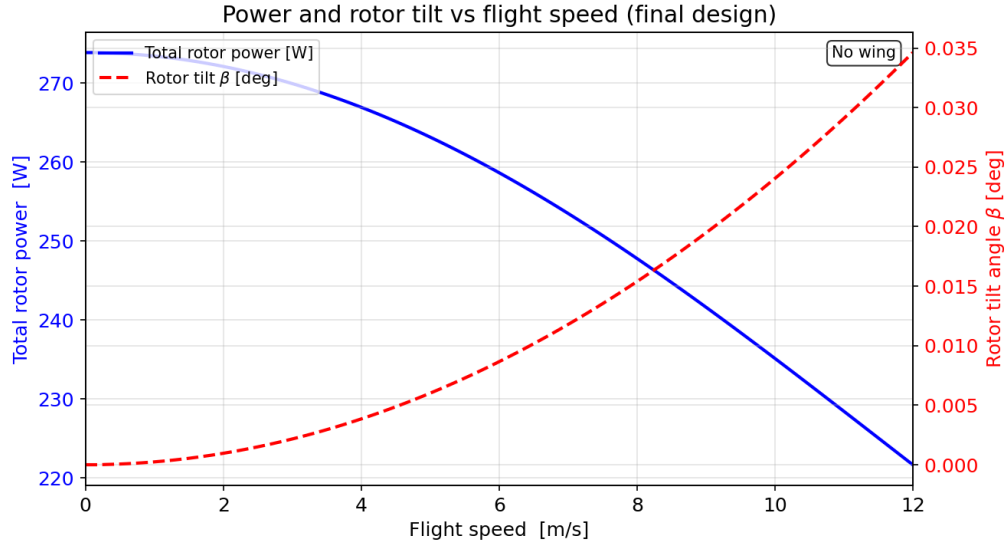


Figure 11: Total rotor power (blue, left axis) and rotor tilt β (red dashed, right axis) vs. flight speed for the final no-wing quadcopter.

Two features follow directly from the theory:

- **Power dips below hover** at low forward speeds. From Equation (75), v_i decreases as V increases, reducing the induced power $\kappa T_{\text{per}}(V \sin \beta + v_i)$ faster than drag rises. This “translational lift” effect is well-known from helicopter aerodynamics.
- **Tilt angle β remains very small** ($< 1^\circ$ at $V = 10 \text{ m/s}$) because at $q = 1.0 \text{ Pa}$ aerodynamic drag is only $\sim 0.03\%$ of the aircraft weight, so $\beta \approx \arctan(D/W) \approx D/W \ll 1$.

6.7 Conclusion

Mars atmospheric density is 60 times lower than Earth’s, giving $q = 1.0 \text{ Pa}$ at $V = 10 \text{ m/s}$. Fixed wings are structurally heavier than the lift they produce for any wing density $\rho_w \geq 7.0 \text{ kg/m}^3$, regardless of planform. The LLT-optimal shape (rectangular, -3° washout, $e > 0.98$) still delivers less than 0.1% power reduction. The final design for forward flight is the bare quadcopter from Tasks 2 and 5, with a forward-flight power of $P = 234.4 \text{ W}$ at $V = 10 \text{ m/s}$.

7 Conclusion

This report developed a full conceptual aerodynamic design for a Mars drone capable of carrying a 2 kg payload, progressing from a first-principles analysis of the Ingenuity reference vehicle through airfoil selection, rotor design, and forward-flight evaluation.

Task 1 established a hover power of approximately 100 W for the Ingenuity coaxial system using momentum theory, providing a validated starting point for the scaling work that followed. Task 2 compared a two-rotor helicopter and a quadcopter, selecting the quadcopter ($N_{\text{rot}} = 4$, $R = 0.53$ m, $N_b = 2$) on the basis of lower power consumption (274.6 W vs. 328.2 W) and longer hover endurance (4.7 min vs. 3.7 min). Task 3 showed that replacing the 2 kg payload with battery cells increases flight time monotonically up to the mass constraint, with the optimal configuration of 42 cells giving approximately 16.3 min.

Task 4 determined that the rotor operates at $Re \approx 1.8 \times 10^4$ — roughly 36 times lower than the equivalent Earth condition — and selected the ROAMX-0201 airfoil for its superior lift performance and stability in this regime. Task 5 refined the rotor design using BEM theory with Prandtl tip-loss corrections, yielding a hover power of approximately 499 W, nearly twice the momentum-theory estimate from Task 2. The difference is primarily attributable to tip losses, which alone account for a $\sim 20\%$ reduction in power when disabled, with the remainder arising from hub geometry and more accurate blade aerodynamic modelling.

Task 6 evaluated fixed-wing integration at $V = 10$ m/s, finding that the extremely low Mars dynamic pressure ($q = 1.0$ Pa) makes wing structural mass outweigh the aerodynamic benefit for any realistic wing material. The LLT-optimal planform (rectangular, $AR = 9$, -3° washout, $e = 0.988$) delivers at most 0.09% power reduction; the final design is therefore the bare quadcopter with a forward-flight power of 234.4 W at $V = 10$ m/s.

Limitations and Further Work

The primary modelling gap across this report is the use of simplified momentum theory in Tasks 2, 3, and 6, while Task 5 demonstrates that the BEM-based power is approximately 225 W higher, showing that the momentum model significantly underestimates the power. Using the BEM baseline in Task 3 would reduce the predicted optimal flight time, and in Task 6 it would raise the absolute power figures while leaving the conclusion - that fixed wings are ineffective on Mars - unchanged. A natural continuation would be to iterate the Task 2 design optimisation using the BEM power model, which may shift the optimal rotor radius and update the Task 3 and Task 6 baselines. Further refinements could include a structural analysis of the rotor blades, more detailed modeling of fuselage and rotor-interference drag, and exploration of variable-pitch control to improve efficiency across the full speed envelope.

8 Code

Code is listed in the appendix [5] as well as linked here: [github](https://github.com/s260073/Fundamentals-of-Aerodynamics-Assignment-3-Group-1)
(<https://github.com/s260073/Fundamentals-of-Aerodynamics-Assignment-3-Group-1>)

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